

Rheumatic fever

- Rheumatic fever is an inflammatory disease that can involve the heart, joints, skin, and brain. It can develop as a complication of inadequately treated streptococcal infection of throat or scarlet fever. Streptococcal infection of throat and scarlet fever are caused by an infection with streptococcus bacteria.
- Rheumatic fever occurs in about 325,000 children each year and about 33.4 million people currently have rheumatic heart disease. Those who develop RF are most often between the ages of 5 and 14, with 20% of first-time attacks occurring in adults.
- The disease is most common in the developing world and among indigenous peoples in the developed world.
- Rheumatic fever can cause permanent damage to the heart, including damaged heart valves and heart failure. Treatments can reduce damage from inflammation, reduces pain and other symptoms and prevent the recurrence of complications

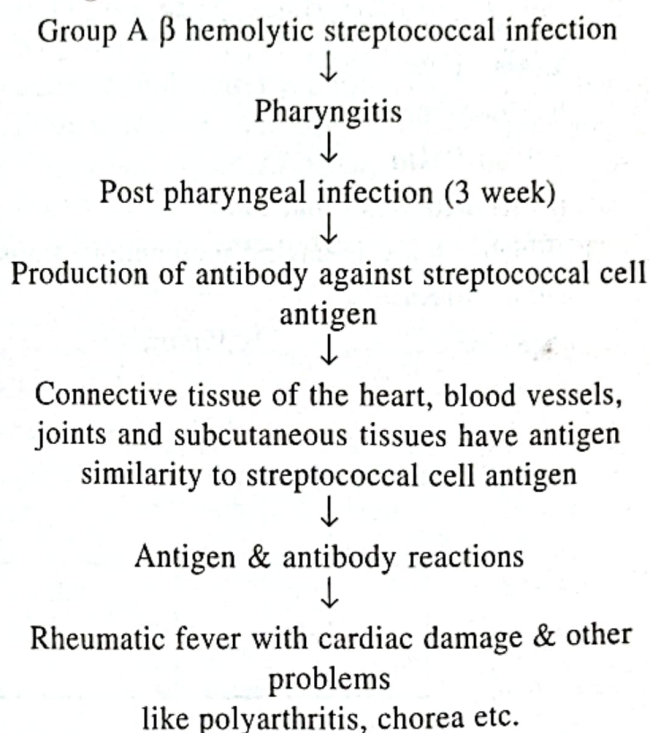
Causes

Rheumatic fever can occur after an infection of the throat with a bacterium called group A streptococcus. Group A streptococcus infections of the throat causes strep throat or, less commonly, scarlet fever. Group A streptococcus infections of the skin or other parts of the body rarely trigger rheumatic fever.

The link between strep infection and rheumatic fever isn't clear, but it appears that the bacterium trigger the immune system. The strep bacterium contains a protein similar to one found in certain tissues of the body. So, immune system cells that would normally target the bacterium may treat the body's own tissues. Particularly tissues of the heart, joints, skin and central nervous system are main

target of such type of infection. This immune system reaction results in inflammation.

Pathogenesis of Rheumatic Fever



Clinical features

Rheumatic fever symptoms vary. Patients have few symptoms or several and symptoms can change during the course of the disease. The onset of rheumatic fever usually occurs about two to four weeks after a streptococcal throat infection. Followings are the common features of rheumatic fever.

- Fever
- Painful and tender joints - most often in the knees, ankles, elbows and wrists
- Pain in one joint that migrates to another joint
- Red, hot or swollen joints
- Small, painless bumps (nodules) beneath the skin
- Chest pain
- Heart murmur
- Fatigue
- Flat or slightly raised, painless rash with a

ragged edge (erythema marginatum)

- Jerky, uncontrollable body movements (Sydenham chorea, or St. Vitus' dance) - most often in the hands, feet and face
- Outbursts of unusual behavior, such as crying or inappropriate laughing, that accompanies Sydenham chorea

Risk factors

Factors that can increase the risk of rheumatic fever include:

- **Family history.** A gene or genes that might make them more likely to develop rheumatic fever.
- **Type of strep bacteria.** Certain strains of streptococcal bacteria are more likely to contribute to rheumatic fever than other strains.
- **Environmental factors.** A greater risk of rheumatic fever is associated with overcrowding, poor sanitation and other conditions.

Complications

Rheumatic fever can cause long-term complications in certain situations. One of the most prevalent complications is rheumatic heart disease. Other heart conditions include:

- Valve stenosis, which is a narrowing of a valve.
- Valve regurgitation, which is a leak in the valve that causes blood to flow in the wrong direction.
- Heart muscle damage, which is an inflammation that can weaken the heart muscle and decrease the heart's ability to pump blood effectively.
- Atrial fibrillation, which is an irregular heart beat in the upper chambers of the heart.
- Heart failure, which happens when the heart can no longer pump blood to all parts of the body.

Diagnostic criteria

Modified Jones criteria were first published in 1944 by T. Duckett Jones, MD. According to revised Jones criteria, the diagnosis of rheumatic fever can be made when two of the major criteria, or one major criterion plus two minor criteria, are

present along with evidence of streptococcal infection: elevated or rising antistreptolysin O titre or DNAase. Signs of a preceding streptococcal infection include: recent scarlet fever, raised antistreptolysin O or other streptococcal antibody titre, or positive throat culture.

Major criteria

- **Polyarthritis:** A temporary migrating inflammation of the large joints, usually starting in the legs and migrating upwards.
- **Carditis:** Inflammation of the heart muscle (myocarditis) which can manifest as congestive heart failure with shortness of breath, pericarditis with a rub, or a new heart murmur.
- **Subcutaneous nodules:** Painless, firm collections of collagen fibers over bones or tendons. They commonly appear on the back of the wrist, the outside elbow, and the front of the knees.
- **Erythema marginatum:** A long-lasting reddish rash that begins on the trunk or arms as macules, which spread outward and clear in the middle to form rings, which continue to spread and coalesce with other rings, ultimately taking on a snake-like appearance. This rash typically spares the face and is made worse with heat.
- **Sydenham's chorea (St. Vitus' dance):** A characteristic series of involuntary rapid movements of the face and arms. This can occur very late in the disease for at least three months from onset of infection.

Minor criteria

- Fever of 38.2–38.9 °C (100.8–102.0°F)
- Arthralgia: Joint pain without swelling (it cannot be included if polyarthritis is present as a major symptom)
- Raised erythrocyte sedimentation rate or C reactive protein
- Leukocytosis
- ECG showing features of heart block, such as a prolonged PR interval (Cannot be included if carditis is present as a major symptom)

- Previous episode of rheumatic fever or inactive heart disease

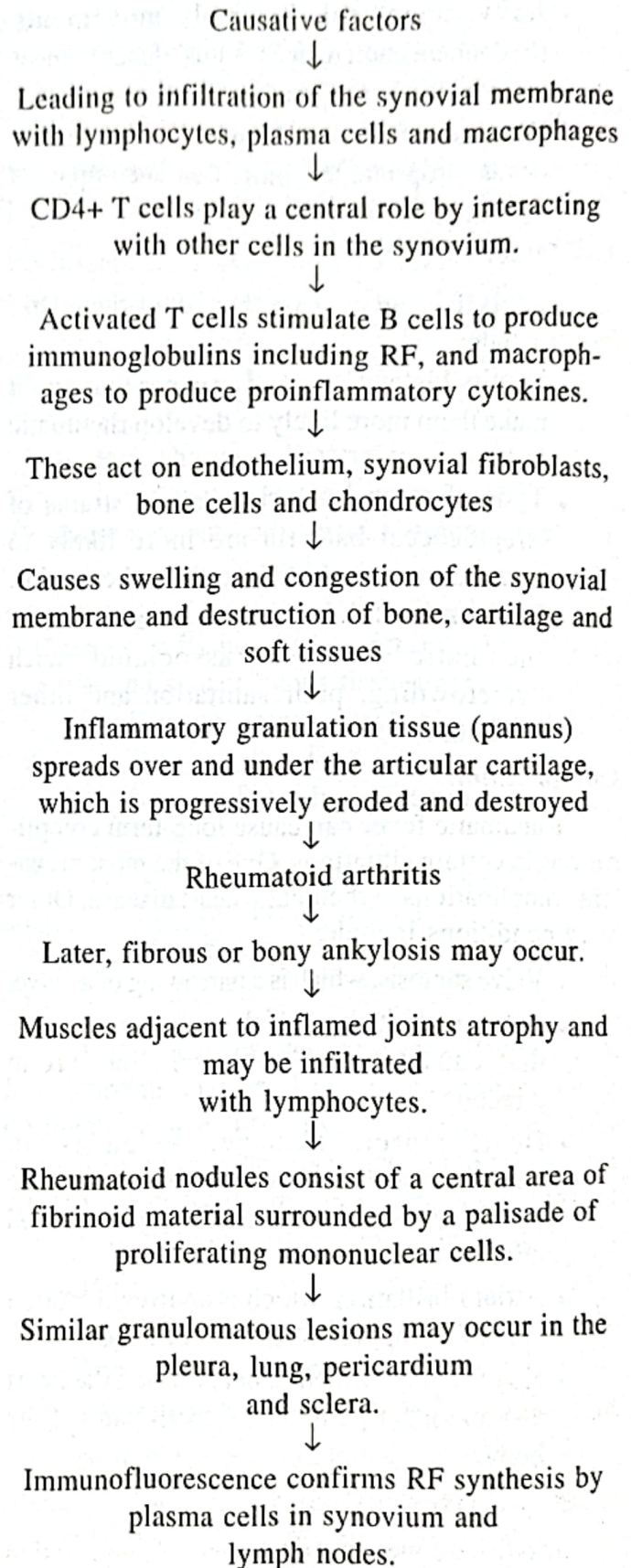
Rheumatoid arthritis (RA)

- Rheumatoid arthritis (RA) is the most common persistent inflammatory arthritis, occurring throughout the world and in all ethnic groups.
- The prevalence is lowest in black Africans and Chinese, and highest in Pima Indians. In Caucasians, it is 1.0-1.5%, with a female to male ratio of 3:7.
- The clinical course is prolonged, with intermittent exacerbations and remissions.
- Factors that associate with a poorer prognosis are disability at presentation, female gender, involvement of MTP joints, smoking and a positive RF and anti-CCP.
- It is hoped that the prognosis of RA will improve as more aggressive early treatment is used but this has yet to be demonstrated.

Etiology

- Both genetic and environmental factors appear to be involved in the pathogenesis of RA.
- The concordance rate of RA is higher in monozygotic (12-15%) than in dizygotic twins (3%), and there is an increased frequency of disease in first-degree relatives of patients.
- 3. Up to 50% of the genetic susceptibility is due to genes in the HLA region.
- 4. HLA-DR4 is the major susceptibility haplotype in most ethnic groups, occurring in 50-75% of Caucasian patients with RA compared to 20-25% of the normal population.
- 5. Although it is thought that RA may be triggered by an infectious agent in a genetically susceptible host, a specific pathogen has not been identified.
- 6. Susceptibility is increased post-partum and by breastfeeding.
- 7. Cigarette smoking is a strong risk factor for developing RA and also associates with greater severity.

Pathophysiology



Clinical features

1. It is characterised by pain, joint swelling and stiffness affecting the small joints of the hands, feet and wrists.

2. Large joint involvement, systemic symptoms and extra-articular features may also occur.

Criteria for diagnosis of rheumatoid arthritis- American rheumatism Association 1988 revision

Diagnosis of RA is made with four or more of the following :

1. Morning stiffness (> 1 hr)
2. Arthritis of three or more joint areas
3. Arthritis of hand joints
4. Symmetrical arthritis
5. Rheumatoid nodules
6. Rheumatoid factor
7. Radiological changes

By convention, symptoms need to have persisted for more than or equal to 6 weeks for the diagnosis of RA.

The 2010 ACR/EULAR classification criteria for RA

According to the ACR/EULAR criteria, patients who should be tested are those (1) who have at least 1 joint with definite clinical synovitis and (2) whose synovitis is not better explained by another disease (lupus, psoriatic arthritis, or gout).

This is a score-based algorithm for RA that incorporates the following 4 factors :

- Joint involvement
- Serology test results
- Acute-phase reactant test results
- Patient self-reporting of the duration of signs and symptoms

The maximum number of points possible is 10. A classification of definitive RA requires a score of 6/10 or higher. Patients with a score lower than 6/10 should be reassessed over time.

A. Joint involvement consists of swelling or tenderness upon examination :

- 1 large joint (ie, shoulders, elbows, hips, knees, ankles) = 0 points
- 2-10 large joints = 1 point
- 1-3 small joints (with or without involvement of large joints), such as MCP, PIP, second to

fifth MTP, thumb interphalangeal (IP), and wrist joints = 2 points

- 4-10 small joints (with or without involvement of large joints) = 3 points
- More than 10 joints (at least 1 small joint, plus any combination of large and additional small joints or joints such as the temporomandibular, acromio-clavicular, or sternoclavicular) = 5 points

B. At least 1 serology test result is needed for RA classification

Points are allocated as follows:

- Negative rheumatoid factor (RF) and negative anti citrullinated protein antibody (ACPA) tested as anti cyclic citrullinated peptide [anti-CCP]) = 0 points
- Low-positive RF or low-positive ACPA = 2 points
- High-positive RF or high-positive ACPA = 3 points

C. At least 1 test acute-phase reactant test result is needed for classification.

- Normal C-reactive protein (CRP) and normal erythrocyte sedimentation rate (ESR) = 0 points
- Abnormal CRP or abnormal ESR = 1 point

D. Points for the patient's self-reporting of the duration of signs or symptoms of synovitis in clinically involved joints are allocated as follows:

- Shorter than 6 weeks = 0 points
- 6 weeks or longer = 1 point

Assessment of disease activity in rheumatoid arthritis

1. Clinical

- Pain
- Visual analogue scale
- Early morning stiffness
- Minutes
- Joint tenderness
- Number of inflamed joints

2. Laboratory

- Acute phase proteins
- Erythrocyte sedimentation rate

- Plasma viscosity
- Presence of anti-CCP antibodies

3. Articular index

- Plain radiography of joints
- USG & MRI of joints

Grading of function in rheumatoid arthritis

I	Fit for all activities	No handicap
II	Moderate restriction	Independent despite handicap and limitation of joint movement
III	Marked restriction	Limited self-care. Some assistance required
IV	Confined to chair or bedbound	Largely incapacitated and dependent

Physical Examination

It is important to assess the following signs :

- Stiffness
- Tenderness
- Pain on motion
- Swelling
- Deformity
- Limitation of motion
- Extra-articular manifestations
- Rheumatoid nodules

Hands

1. 'Swan neck' deformity
2. The boutonniere or 'button hole' deformity
3. Z deformity of the thumb
4. Dorsal subluxation of the ulna at the distal radio-ulnar joint is common

Foot

1. Dorsal subluxation of the MTP joints may result in 'cock-up' toe deformities.
2. In the hindfoot, calcaneovalgus (eversion) is the most common deformity, reflecting damage to the ankle and subtalar joint.
3. Popliteal (Baker's) cysts usually occur in combination with knee synovitis.
4. Rupture, often induced by knee flexion in the presence of a large effusion, leads to calf pain and swelling.

Investigations

1. ESR and CRP are usually raised (acute phase response)
2. Rheumatoid Factor and anti-CCP antibodies
3. Plain X-rays of the hands, wrist and feet are useful
4. Peri articular osteoporosis is common during the early stage of disease
5. Non-proliferative marginal joint erosions on X-ray are characteristic but uncommon within the first year.
6. Ultrasound and MRI are more sensitive than X-rays at detecting early erosions.
7. Patients who are suspected of having atlanto-axial disease should have lateral X-rays taken in flexion and extension, and the degree of cord compression should be established with MRI.
8. In patients with Baker's cyst, Doppler ultrasound and an arteriogram may be required to establish the diagnosis, since deep Venous thrombosis (DVT) and Baker's cyst may coexist.

Outcome measures in rheumatoid arthritis

- Functional
- Grip strength
- Activities of daily living
- Functional index
- Health Status Questionnaire

Dimensions	Subdimensions	Components (e.g.)
Death	Upper lower	Grip, feed, walk, climb
Disability	Physical	Pain, fatigue,
Discomfort	Psychological	Depression, anxiety
Iatrogenic	Medical	Dyspepsia, haemorrhage
	Surgical	Prosthesis infection and loosening
Economic	Direct	Drugs, consultations,
	Indirect	Loss of work, social

Systemic Lupus Erythematosus (SLE)

- Systemic lupus erythematosus (SLE) is an autoimmune disorder characterized by antibodies to nuclear and cytoplasmic antigens, multisystem inflammation, protean clinical manifestations, and a relapsing and remitting course.
- More than 90% of cases of SLE occur in women, frequently starting at childbearing age.

Pathophysiology and Etiology

- SLE is an autoimmune disorder characterized by multisystem inflammation with the generation of autoantibodies.
- Although the specific cause of SLE is unknown
- Multiple factors are associated with the development of the disease, including genetic, epigenetic, ethnic, immunoregulatory, hormonal, and environmental factors.

Clinical presentation-Signs and symptoms of SLE

- SLE is a chronic inflammatory disease that can affect almost any organ system, although it mainly involves the skin, joints, kidneys, blood cells, and nervous system.
- Its presentation and course are highly variable, ranging from indolent to fulminant.

In childhood-onset SLE, the following clinical manifestations are more commonly found than in adults :

- Malar rash
- Ulcers/mucocutaneous involvement
- Kidney involvement, proteinuria, urinary cellular casts
- Seizures
- Thrombocytopenia
- Hemolytic anemia
- Fever
- Lymphadenopathy.

In adults, Raynaud pleuritis and sicca are twice as common as in children and adolescents.

- The classic presentation of a triad of fever, joint pain, and rash in a woman of childbearing

age should prompt investigation into the diagnosis of SLE.

Patients may present with any of the following manifestations:

- Constitutional (eg, fatigue, fever, arthralgia, weight changes)
- Musculoskeletal (eg, arthralgia, arthropathy, myalgia, frank arthritis, avascular necrosis)
- Dermatologic (eg, malar rash, photosensitivity, discoid lupus)
- Renal (eg, acute or chronic kidney failure, acute nephritic disease)
- Neuropsychiatric (eg, seizure, psychosis)
- Pulmonary (eg, pleurisy, pleural effusion, pneumonitis, pulmonary hypertension, interstitial lung disease)
- Gastrointestinal (eg, nausea, dyspepsia, abdominal pain)
- Cardiac (eg, pericarditis, myocarditis)
- Hematologic (eg, cytopenias such as leukopenia, lymphopenia, anemia, or thrombocytopenia)

Note- In patients with suggestive clinical findings, a family history of autoimmune disease should raise further suspicion of SLE.

Ancillary investigations & Diagnosis of SLE

- The diagnosis of SLE is based on a combination of clinical findings and laboratory evidence.
- Familiarity with the diagnostic criteria helps clinicians to recognize SLE and to subclassify this complex disease based on the pattern of target-organ manifestations.
- The American College of Rheumatology (ACR) and the European League Against Rheumatism (EULAR) published new criteria for the classification of SLE in 2019. These criteria represent current concepts of SLE, and have excellent specificity and sensitivity.

The American College of Rheumatology (ACR) and the European League Against Rheumatism (EULAR)

- The EULAR/ACR classification requires an

antinuclear antibody (ANA) titer of at least 1:80 on HEP-2 cells or an equivalent positive test at least once; otherwise, the patient is considered not to have SLE.

- If it is present, 22 "additive weighted" classification criteria are considered, comprising seven clinical domains and three immunologic domains.
- Each criterion is assigned points, ranging from 2 to 10. Patients with at least one clinical criterion and 10 or more points are classified as having SLE.

EULAR/ACR Clinical Domains and Criteria for SLE

Domain	Criteria	Points
Constitutional	Fever	2
Hematologic	Leukopenia	3
	Thrombocytopenia	4
	Autoimmune hemolysis	4
Neuropsychiatric	Delirium	2
	Psychosis	3
	Seizure	5
Mucocutaneous	Non-scarring alopecia	2
	Oral ulcers	2
	Subacute cutaneous or discoid lupus	4
	Acute cutaneous lupus	6
Serosal	Pleural or pericardial effusion	5
	Acute pericarditis	6
Musculoskeletal	Joint involvement	6
Renal	Proteinuria > 0.5 g/24 h	4
	Kidney biopsy class II or V lupus nephritis	8
	Kidney biopsy class III	10
	or IV lupus nephritis	

Note the following :

- A criterion should not be counted if there is a more likely explanation for it than SLE
- Occurrence of a criterion on at least one occasion is sufficient
- Criteria need not occur simultaneously

- Within each domain, only the highest-weighted criterion is counted toward the total score

Testing

Laboratory studies used in the diagnosis of SLE are as follows:

- CBC with differential
- Serum creatinine
- Urinalysis with microscopy
- ESR or CRP level
- Complement levels
- Liver function tests
- Creatine kinase assay
- Spot protein/spot creatinine ratio
- Autoantibody tests

Autoantibody tests

Autoantibody Tests for SLE

Test	Description
ANA	Screening test; sensitivity 95%; not diagnostic without clinical features
Anti-dsDNA	High specificity; sensitivity only 70%; level is variable based on disease activity
Anti-Sm	Most specific antibody for SLE; only 30-40% sensitivity
Anti-SSA (Ro) or Anti-SSB (La)	Present in 15% of patients with SLE and other connective-tissue diseases such as Sjogren syndrome; associated with neonatal lupus
Anti-ribosomal P	Uncommon antibodies that may correlate with risk for CNS disease, including increased hazards of psychosis in a large inception cohort, although the exact role in clinical diagnosis is debated.
Anti-RNP	Included with anti-Sm, SSA, and SSB in the ENA profile; may indicate mixed connective-tissue disease with overlap SLE, scleroderma, and myositis
Anticardiolipin	IgG/IgM variants measured with

	ELISA are among the antiphospholipid antibodies used to screen for antiphospholipid antibody syndrome and pertinent in SLE diagnosis
Lupus anticoagulant	Multiple tests (eg, direct Russell viper venom test) to screen for inhibitors in the clotting cascade in antiphospholipid antibody syndrome
Direct Coombs test	Coombs test-positive anemia to denote antibodies on RBCs
Anti-histone	Drug-induced lupus ANA antibodies are often of this type (eg, with procainamide or hydralazine; p-ANCA-positive in minocycline-induced drug-induced lupus)
ANA = antinuclear antibody; CNS = central nervous system; ds-DNA = double-stranded DNA; ELISA = enzyme-linked immunoassay; ENA = extractable nuclear antigen; Ig = immunoglobulin; p-ANCA = perinuclear antineutrophil cytoplasmic antibody; RBCs = red blood cells; RNP = ribonucleic protein; SLE = systemic lupus erythematosus; Sm = Smith; SSA = Sjogren syndrome A; SSB = Sjogren syndrome B.	

Imaging studies

The following imaging studies may be used to evaluate patients with suspected SLE:

- Joint radiography
- Chest radiography and chest CT scanning
- Echocardiography
- Brain MRI/MRA
- Cardiac MRI

Procedures

Procedures that may be performed in patients with suspected SLE include the following:

- Arthrocentesis
- Lumbar puncture
- Kidney biopsy

Ankylosing spondylitis

1. Ankylosing spondylitis (AS) is characterised by a chronic inflammatory arthritis predominantly affecting the sacroiliac joints

and spine, which can progress to bony fusion of the spine.

2. It has a peak onset in the second and third decades
3. Male female ratio of about 3:1.
4. In Europe, more than 90% of those affected are HLA-B27-positive.
5. The overall prevalence is less than 0.5% in most communities.
6. Over 75% of patients are able to remain in employment and enjoy a good quality of life.
7. Even if severe ankylosis develops, functional limitation may not be marked as long as the spine is fused in an erect posture

Etiopathogenesis

It result from exposure to a common environmental pathogen in genetically susceptible individuals



Although no specific trigger has been identified



Chronic prostatitis is more common than expected but appears to be non-infective in origin



Increased faecal carriage of *Klebsiella aerogenes* occurs in patients with established AS



May relate to exacerbation of both joint and eye disease.

Clinical features

1. The onset is usually insidious, with recurring episodes of low back pain and marked stiffness.
2. Radiation of pain to the buttocks or posterior thighs may be misdiagnosed as sciatica.
3. Unlike mechanical back pain, symptoms extend over many segments and are axial and symmetrical in distribution.
4. Symptoms are most marked in the early morning and after inactivity
5. Symptoms are relieved by movement.
6. Although the lumbosacral area is usually the first and worst affected region

7. Some patients present with mainly thoracic or neck symptoms.
8. The disease tends to ascend the spine slowly and eventually
9. After several years, the whole spine may be affected.
10. As the spine becomes progressively ankylosed, spinal rigidity and secondary osteoporosis predispose to spinal fracture.
11. Spinal cord compression is rare.

Physical examination findings initially

1. Failure to obliterate the lumbar lordosis on forward flexion
2. Restriction of movements of the lumbar spine in all directions
3. Possible pain on sacroiliac stressing

Physical examination findings of progressive disease

1. Stiffness increases throughout the spine
2. Chest expansion frequently becomes restricted.
3. Spinal fusion varies in its extent and in most cases does not cause a gross flexion deformity.
4. A few cases develop marked kyphosis of the dorsal and cervical spine that may interfere with forward vision.
5. This may prove incapacitating, especially when associated with fixed flexion contractures of hips or knees.
6. Pleuritic chest pain aggravated by breathing is common and results from costovertebral joint involvement.
7. Plantar fasciitis, Achilles tendinitis and tenderness over bony prominences such as the iliac crest and greater trochanter are common, reflecting inflammation at the sites of tendon insertions (enteritis).

Peripheral arthritis

1. Up to 40% of patients have peripheral arthritis.
2. It is usually asymmetrical at first, mainly affecting hips, knees, ankles or shoulders.
3. Involvement of a peripheral joint, most commonly ankle, knee or elbow, may precede

the development of spinal symptoms in around 10% of cases.

4. In a further 10%, symptoms begin in childhood as one variety of pauciarticular juvenile idiopathic arthritis.

Extra-articular disease

1. Fatigue is often a major complaint and may result from both chronic interruption of sleep due to pain, as well as chronic systemic inflammation.
2. Acute anterior uveitis is the most common extra-articular feature.
3. Conjunctivitis
4. Prostatitis : usually asymptomatic
5. Aortic incompetence
6. Mitral incompetence
7. Cardiac conduction defects
8. Pericarditis
9. Amyloidosis
10. Atypical upper lobe pulmonary fibrosis

Investigations

A. Blood

1. The ESR and CRP are usually raised in active disease.
2. RF and other autoantibodies are usually negative.
3. Testing for HLA-B27 can be a helpful investigation for pauciarticular juvenile idiopathic arthritis but is unhelpful in adults with spinal symptoms.

B. X-ray

1. X-ray changes are characteristic but may take year to develop.
2. Sacroiliitis is often the first abnormality beginning in the lower synovial parts of the joints with irregularity and loss of cortical margins, widening of the joint space and subsequently sclerosis, narrowing and fusion.
3. Lateral thoracolumbar spine X-rays may show anterior 'squaring' of vertebrae due to erosion and sclerosis of the anterior corners and periostitis of the waist.

4. Bridging syndesmophytes are fine and symmetrical and follow the outermost fibres of the annulus.
5. Ossification of the anterior longitudinal ligament and, facet joint fusion may also be visible and may result in the typical 'bamboo' spine
6. Erosive changes may be seen in the symphysis pubis, the ischial tuberosities and peripheral joints.
7. Osteoporosis and atlanto-axial dislocation can occur as late features.

C. MRI

MRI is more sensitive for detection of early sacroiliitis, but is seldom required.

Gout

1. Gout is a true crystal deposition disease, and is defined as the pathological reaction of the joint or periarticular tissues to the presence of monosodium urate monohydrate (MSU) crystals.
2. It results from an increased body pool of urate with hyperuricemia.
3. It typically is characterized by episodic acute and chronic arthritis caused by deposition of MSU crystals in joints and connective tissue tophi and the risk for deposition in kidney interstitium or uric acid nephrolithiasis.

Epidemiology

- The prevalence of gout varies between populations but is approximately 1-2%, with a strong male predominance (> 5:1).
- It is the most common inflammatory arthritis in men and in older women.
- Prevalence increases with increasing serum uric acid (SUA) and with age.
- The incidence and prevalence of gout have increased in many countries in recent decades, in parallel with increased longevity and the higher prevalence of metabolic syndrome, of which hyperuricaemia is an integral component.

Hyperuricaemia

- It is defined as an SUA level greater than 2

standard deviations above the mean for the population.

- Only a minority of hyperuricaemic people develop gout, emphasising the importance of tissue factors in crystal formation.

Aetiology Primary gout

- About one-third of the body uric acid pool is derived from dietary sources and two-thirds from endogenous purine metabolism
- The concentration of uric acid in body fluids depends on the balance between its synthesis and elimination by the kidneys (two-thirds) and gut (one-third).
- Purine nucleotide synthesis and degradation are regulated by a network of enzyme pathways; xanthine oxidase catalyses the end conversion of hypoxanthine to xanthine and then xanthine to uric acid.
- In over 90% of patients with primary gout, hyperuricaemia results from an inherited defect in fractional uric acid excretion which impairs their ability to increase urate excretion in response to a purine load ('underexcretors').

Risk factors and associations for primary gout include

- Metabolic syndrome, high alcohol intake (predominantly beer which contains guanosine), and diets relatively high in red meat or fructose or relatively low in vitamin C or coffee.

Secondary gout

- Secondary gout results from hyperuricaemia due to renal impairment or chronic diuretic use.

Factors that predispose to chronic hyperuricaemia and gout

Diminished renal excretion (common)

- Inherited isolated renal tubular defect ('under-excretors')
- Renal failure
- Chronic drug therapy
 - Thiazide and loop diuretics
 - Low-dose aspirin Ciclosporin
 - Pyrazinamide

- Lead toxicity (e.g. in 'moonshine' drinkers)
- Lactic acidosis (alcohol)

Increased production of uric acid (uncommon)

- Increased purine turnover
Chronic myeloproliferative or lymphoproliferative disorders (e.g. polycythaemia, chronic lymphatic leukaemia)
- Increased de novo synthesis ('over-producers')
- Unidentified abnormality (most common)
- Specific enzyme defect (rare)
- Hypoxanthine-guanine phosphoribosyl transferase deficiency
- Phosphoribosyl pyrophosphate synthetase over-activity
- Glucose-6-phosphatase deficiency

Acute and Chronic Arthritis

1. Acute arthritis is the most common early clinical manifestation of gout
2. Usually, only one joint is affected initially, but polyarticular acute gout can occur in subsequent episodes.
3. Joints involvement
 - The first MTP joint is affected in over 50% of cases 'podagra' ('seizing the foot').
 - Other common sites are, in order of decreasing frequency, the ankle, midfoot, knee, small joints of hands, wrist and elbow.
 - The axial skeleton and large proximal joints are rarely involved and never as the first site.
 - **Typical attacks have the following characteristics :**
 - Extremely rapid onset, reaching maximum severity in just 2-6 hours, often waking the patient in the early morning severe pain, often described as the 'Worst pain ever'
 - Extreme tenderness: the patient is unable to wear a socks or to let bedding rest on the joint
 - marked swelling with overlying red, shiny skin
 - Self-limiting over 5-14 days, with complete return to normality.
4. First episode
 - The first episode of acute gouty arthritis frequently begins at night with dramatic joint

pain and swelling.

- Joints rapidly become warm, red, and tender, with a clinical appearance that often mimics that of cellulitis.
 - Early attacks tend to subside spontaneously within 3-10 days, and most patients have intervals of varying length with no residual symptoms until the next episode.
5. Precipitating factors
Dietary excess, trauma, surgery, excessive ethanol ingestion, hypouricemic therapy, and serious medical illnesses such as myocardial infarction and stroke
 6. Heberden's or Bouchard's nodes
 - Inflamed Heberden's or Bouchard's nodes may be a first manifestation of gouty arthritis
 7. Chronic nonsymmetric synovitis
 - After many acute mono- or oligoarticular attacks, a proportion of gouty patients may present with a chronic nonsymmetric synovitis, causing potential confusion with rheumatoid arthritis
 - Less commonly, chronic gouty arthritis will be the only manifestation, and, more rarely, the disease will manifest only as periarticular tophaceous deposits in the absence of synovitis.
 - Women represent only 5-20% of all patients with gout.
 - Premenopausal gout is rare; it is seen mostly in individuals with a strong family history of gout.

Laboratory Diagnosis

1. Definitive diagnosis requires identification of MSU crystals in the aspirate from a joint, bursa or tophus
2. During acute gouty attacks, needle-shaped MSU crystals typically are seen both intracellularly and extracellularly.
3. Synovial fluid leukocyte counts are elevated from 2000 to 60,000/L.
4. Effusions appear cloudy due to the increased numbers of leukocytes.
5. Large amounts of crystals occasionally produce a thick pasty or chalky joint fluid.

6. Arthrocentesis of these joints is a useful technique to establish the diagnosis of gout between attacks.
 7. Serum uric acid levels can be normal or low at the time of an acute attack, as inflammatory cytokines can be uricosuric and effective initiation of hypouricemic therapy can precipitate attacks.
 8. A 24-h urine collection for uric acid can, in some cases, be useful in assessing the risk of stones, elucidating overproduction or underexcretion of uric acid, and deciding whether it may be appropriate to use a uricosuric therapy.
 - Excretion of >800 mg of uric acid per 24 h on a regular diet suggests that causes of overproduction of purine should be considered.
 9. Other tests
 - Urinalysis, serum creatinine, hemoglobin, white blood cell (WBC) count, liver function tests, and serum lipids should be obtained because of possible pathologic sequelae of gout
 10. Radiographic Features
 - Cystic changes, well-defined erosions with sclerotic margins (often with overhanging bony edges), and soft tissue masses are characteristic features of advanced chronic tophaceous gout.
 - Ultrasound, CT and MRI are being studied and are likely to become more sensitive for early changes.
- b. Reactive arthritis
 - c. Rheumatoid arthritis (RA)
 - d. Ankylosing spondylitis
 2. During acute attacks of which disease needle-shaped MSU crystals typically are seen both intracellularly and extracellularly
 - a. Gouty arthritis
 - b. Reactive arthritis
 - c. Psoriatic arthritis
 - d. Ankylosing spondylitis
 3. Antinuclear antibodies (ANA) are positive in >98% of patients during the course of disease and High-titer IgG antibodies to double-stranded DNA and antibodies to the Sm antigen are both specific for
 - a. SLE and, therefore, favor the diagnosis in the presence of compatible clinical manifestations
 - b. RA and, therefore, favor the diagnosis in the presence of compatible clinical manifestations
 - c. Gouty arthritis and, therefore, favor the diagnosis in the presence of compatible clinical manifestations
 - d. Reactive arthritis and, therefore, favor the diagnosis in the presence of compatible clinical manifestations
 4. Inflamed Heberden's or Bouchard's nodes may be a first manifestation of which arthritis
 - a. Gout
 - b. Osteoarthritis
 - c. Rheumatoid arthritis
 - d. Rheumatic fever
 5. Foot deformities (such as hammertoes or clawtoes and Charcot foot) are seen in
 - a. Rheumatic fever
 - b. Rheumatoid arthritis
 - c. Diabetes mellitus (DM)
 - d. Ankylosing spondylitis

MCQs

1. Serum antibodies to cyclic citrullinated peptides (anti-CCPs) are now recognized to be a valuable biomarker of diagnostic and prognostic significance of
 - a. Gouty arthritis

Answer

1.	c	2.	a	3.	a
4.	a	5.	b		